MIL-C-24903A(CG)
6 December 1984
SUPERSEDING
MIL-C-24903(CG)
23 April 1982

MILITARY SPECIFICATION

CLOTH, ELASTIQUE, POLYESTER/WOOL

This specification is approved for use by the Navy Clothing and Textile Research Facility, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of water-repellent treated polyester and wool elastique cloth.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified.

SPECIFICATIONS

FEDERAL

PPP-P-1132 - Packaging of Woolen; Worsted and Wool Blend (Synthetic Fiber; Cotton) Fabrics

MILITARY

MIL-C-43665 - Cloth Wool: Mothproofing Treatment of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Officer in Charge, Navy Clothing and Textile Research Facility, 21 Strathmore Road, Natick, MA 01760-2490 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8305

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STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-655 - Provisions for Evaluating Quality of Cloth, Wool, Worsted and Wool Blends

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

LAWS AND REGULATIONS

U.S. DEPARTMENT OF AGRICULTURE

Methods of Test for Grades of Wool Top

(Applications for copies should be addressed to the U.S. Department of Agriculture, Agricultural Marketing Service, Washington, DC 20402).

Rules and Regulations Under the Wool Products Labeling Act of 1939

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D 1424 - Tear Resistance of Woven Fabrics by Falling-Pendulum (Elmendorf) Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Standard sample. The finished cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.3 and 6.2).

3.3 Material.

3.3.1 Fibers (see 6.5).

3.3.1.1 Wool. The stock shall be fleece or pulled wool or both, not lower in grade than 64's U.S. Standard, when tested as specified in 4.4.1.2.

3.3.1.2 Polyester. The stock shall be semidull polyester fiber made from polyethylene glycol terephthalate either homo polymer or modified polymer as appropriate (see 4.4.1.1). The denier shall be suitable to blend adequately with the wool grade to meet the requirements of this specification.

3.3.2 Yarns. The polyester and wool fiber shall be blended combed, and spun into a two ply yarn for both the warp and filling. The finished cloth shall consist of the following fiber percentages based on the oven-dry weight when tested as specified in 4.6.

55 percent (min), 60 percent (max) polyester fiber

40 percent (min), wool fiber, 64's U.S. Standard (or finer)

NOTE: Selvage yarns: To prevent edges of the cloth from building up when rolled, the selvage yarns may be made of a finer count than the body of the cloth. Also the fiber content of the selvage yarns shall be the same as that used in the body yarn.

3.4 Physical requirements. The finished cloth shall conform to the requirements shown in Table I when tested as specified in 4.6.

TABLE I. - Physical requirements

Ounces per linear yard 60 inches (Grams per linear m - 1.52 m wide) Min.	Yarns per inch (2.54 cm) Minimum		Breaking Strength in Pounds (Newton's) Minimum		Tearing Strength in Pounds (Newton's) Minimum	
	Warp	Filling	Warp	Filling	Warp	Filling
16.5 (512)	120	85	220 (979)	145 (645)	4.0 (17.8)	3.0 (13.4)

3.4.1 Weave. The weave shall be a right hand twill as shown in Figure 1, when tested as specified in 4.6.

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3.5 Performance.

3.5.1 Dimensional stability. The finished cloth shall not shrink or elongate more than 2.0 percent in the direction of the warp or filling when tested as specified in 4.6.

3.5.2 pH. The pH value of the water extraction of the finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.6.

3.6 Length and put-up. Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths each not less than 50 yards (45.7 m). Each length shall be put-up on full width rolls as specified in PPP-P-1132.

3.7 Width. Unless otherwise specified (see 6.2), the width shall be no less than 60 inches (1.52 m) exclusive of secure non-raveling selvages or exclusive of tuck in when woven on sulzer-type looms.

3.8 Mothproofing. Mothproofing of the wool material shall be in accordance with MIL-C-43665.

3.9 Color. The color shall be Blue 3362 and shall match the standard sample. The color shall be obtained by piece dyeing.

3.9.1 Matching. The color of the finished cloth shall match the standard sample under artificial daylight having a color temperature of 7000 ± 500 Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2850 ± 100 Kelvin.

3.9.2 Colorfastness. The finished cloth shall show fastness to wet drycleaning, crocking, perspiration and light equal to or better than the standard sample. When no standard sample has been established or designated as applicable to colorfastness, the finished cloth shall show "good" fastness to wet drycleaning, perspiration, and light, and shall show a Munsell Value for crocking not lower than 8.5. The fastness to light exposure time for good fastness shall be 40 standard fading hours. Test shall be as specified in 4.6.

3.10 Finish. The cloth shall be scoured, brushed, sheared, pressed, decated, and treated with an approved fluorocarbon type water repellent treatment (see 6.7) so as to provide a finish equal to that of the standard sample.

3.10.1 Water repellency. The finished cloth shall conform to the requirements in Table II and 3.10.2 when tested as specified in 4.6.

Table II - Water repellency requirements

	Hydrostatic height min. centimeters	Dynamic absorption maximum
Initial	25	15%
After one dry cleaning	25	--

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3.10.2 Spray rating. The results of the three individual determinations on the sample unit for spray rating shall be equal to or better than ratings 90, 90, 90 initial and 80, 80, 80 after one dry cleaning when tested as specified in 4.6.

3.11 Resistance to organic liquid. The finished cloth shall show no wetting by n-tetradecane initially as specified in 4.6.

3.12 Wool content label. The cloth shall be labeled in accordance with the Wool Product Labeling Act of 1939.

3.13 Marking. The back of the cloth shall be marked "BACK" at both ends of each roll with letters not less than 1/2 inch (1.3cm) in height (see 4.4.2.3). The marking shall be clearly legible in any indelible marking medium used commercially.

3.14 Workmanship. The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards (83.6 Sq.m) when calculated as specified in Section 4 shall not exceed the established maximum point value. Each defect shall be marked with a 1 to 1-1/2 inch (2.5 to 3.8cm) long string. The string shall be sewn into the selvage near the defect. A red string shall represent a three or four point defect and a white string shall represent a one or two point defect (see 4.4.2.5).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), the first article of the finished cloth shall be visually examined for color, finish, and appearance and shall be tested for chemical and physical properties in accordance with the applicable methods specified in 4.6.

4.4 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-655, except where otherwise indicated.

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4.4.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.4.1.1 Polyester. The contractor shall furnish a certificate of compliance for the polyester fiber requirements of identification specified in 3.3.1.2.

4.4.1.2 Wool. The wool shall be examined for grade (see 3.3.1.1) in undyed top form by comparison with the applicable U.S. Standard (see 2.1.2). In the event of a dispute, the wool grade shall be determined by the width method (wedge). The sample unit shall be 1 yard (0.91 m) of undyed top. The sample size (number of sample units) shall be as shown below. The lot size shall be expressed in units of 1 pound (.454 kg). The results from each sample unit shall be reported separately. The lot shall be unacceptable if a single sample unit fails to meet the required grade.

Lot size-pounds (kg)	Sample Size
800 or less (363 or less)	2
801 up to and including 22,000 (363 up to and including 9988)	3
22,001 and over (9989 and over)	5

4.4.1.3 In-process inspection. Inspection shall be made at any point, or during any phase, of the manufacturing process to assure that only approved components are used and that the dyeing requirements have been adhered to (see 3.9 and 3.10). The government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

4.4.2 Examination of the end item. Examination of the end item shall be in accordance with 4.4.2.1 through 4.4.2.5.

4.4.2.1 Yard-by-yard examination. The yard-by-yard examination shall be as specified in MIL-STD-655. The fabric quality shall be grade 10.

4.4.2.1.1 Limits of slubs and knots. Only slubs and knots which exceed "D" for slubs and "C" for knots on the Sears Fabric Defect Scale (see 6.4) shall be scored.

4.4.2.2 Examination for length. The examination for length shall be as specified in MIL-STD-655.

4.4.2.3 Examination for back marking and compliance with the Wool Products Labeling Act. The examination for back marking and compliance with the Wool Products Labeling Act (see 2.1.2) shall be as specified in MIL-STD-655.

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4.4.2.4 Examination for shade match and finish, individual rolls. Each roll in the lot shall be examined visually for shade match (see 3.9) and finish (see 3.10). A roll shall be unacceptable if it fails to meet the requirements for shade match or finish. The sample unit shall be a 4 inch by 20 inch (10.2 cm x 50.8 cm) swatch of the cloth. A sample unit shall be drawn from each roll in the lot.

4.4.2.5 Examination for defect marking (stringing of defects). The examination for defect marking shall be as specified in MIL-STD-655.

4.5 Examination of packaging requirements. Examination shall be made in accordance with the provisions of PPP-P-1132 to determine that packaging, packing, and marking comply with the Section 5 requirements.

4.6 Tests. The methods of testing specified in FED-STD-191, wherever applicable and as listed in Table III shall be followed. The physical and chemical values specified in Section 3 apply to the result of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 5 continuous yards (4.55 continuous meters) full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The lot size shall be expressed in units of 1 yard (0.91 m). All test reports shall contian the individual values utilized in expressing the final result. The sample size (number of sample units) shall be in accordance with the following:

Lot size-yards (meters)	Sample size
800 or less (728 or less)	2
801 up to and including 22,000 (729 up to and including 20,020)	3
22,001 and over (20,021 and over)	5

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Table III. Test methods

Characteristic	Requirement paragraph	Test Method
Yarn ply	3.3.2	Visual
Fiber content:		
Polyester (percent)	3.3.2	2102 1/
Wool (percent)	3.3.2	2102 1/
Colorfastness:		
Wet drycleaning	3.9.2	5621 & 5622
Crocking	3.9.2	5651
Light	3.9.2	5660
Perspiration	3.9.2	5680
Weight	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Tearing strength:		
Warp	3.4	ASTM-D 1424-63 2/
Filling	3.4	ASTM-D 1424-63 2/
Dimensional stability:		
Warp	3.5.1	5558
Filling	3.5.1	5558
Weave	3.4.1	Visual 3/
pH	3.5.2	2811
Hydrostatic pressure:		
Initial	3.10.1	5514
After 1 drycleaning	3.10.1	5508 & 5514
Dynamic absorption:		
Initial	3.10.1	5500
Spray rating:		
Initial	3.10.2	5526
After 1 drycleaning	3.10.2	5508 & 5526
Resistance to organic liquid:		
Initial	3.11	4.6.1
Mothproofing	3.8	4/

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1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirements.

2/ Standard instrument with one augmenting weight.

3/ One determination per sample unit and the results reported as "pass" or "fail".

4/ As tested in MIL-C-43665.

4.6.1 Test for resistance to organic liquid. Place a small specimen of the cloth on a smooth horizontal surface, face up. Using a pipette or eye dropper, gently deposit one drop of n-tetradecane on the surface of the specimen. After one minute, examine the specimen under light at an angle. Absence of light reflectance at the fabric-drop interface shall be taken as evidence of wetting. Three specimens (or areas) taken at various locations across the sample shall be tested. Evidence of wetting on one or more specimens shall be considered cause for rejection of the lot represented by the sample.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A and Commercial. The cloth shall be put up and packaged in accordance with the applicable requirements of PPP-P-1132.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B, or Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1132.

5.3 Marking. In addition to any special marking required in the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1132.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use in the manufacture of the Coast Guard men's and women's overcoats.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. When a first article inspection is required (see 3.2). The item will be tested and should be a first article sample. The contracting officer should include specific instructions, quantity, and testing and approval of the first article.

c. Length required if other than specified (see 3.6).

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d. Width of cloth required when other than specified (see 3.7).

e. Selection of applicable levels of put-up, packaging and packing (see 5.1 and 5.2).

6.3 Standard sample. For access to standard sample (see 3.1), address the procuring office issuing the invitation for bids.

6.4 Sears Fabric Defect Scales are available from Sears, Roebuck and Company, Department 817 (ATTN: BSC 23-29), "Sears Defect Replica Kit", Sears Tower, Chicago, IL 60684.

6.5 Recycled material. The use of recycled material is not permitted.

* 6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.7 Approval of fluorocarbon water repellent. Approval of the fluorocarbon type water repellent treatment is the responsiblity of the US Army Natick Research and Development Center (GL), Natick, MA 01760 and is based on extensive tests, including those for toxicity which are not set forth in this document. Because of the time necessary to conduct full evaluation, only those chemical treatments already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

Custodian:
Navy - NU

Preparing Activity:
Navy - NU

Review Activity:
DLA - CT
Navy - CG

Project No. 8305-N007

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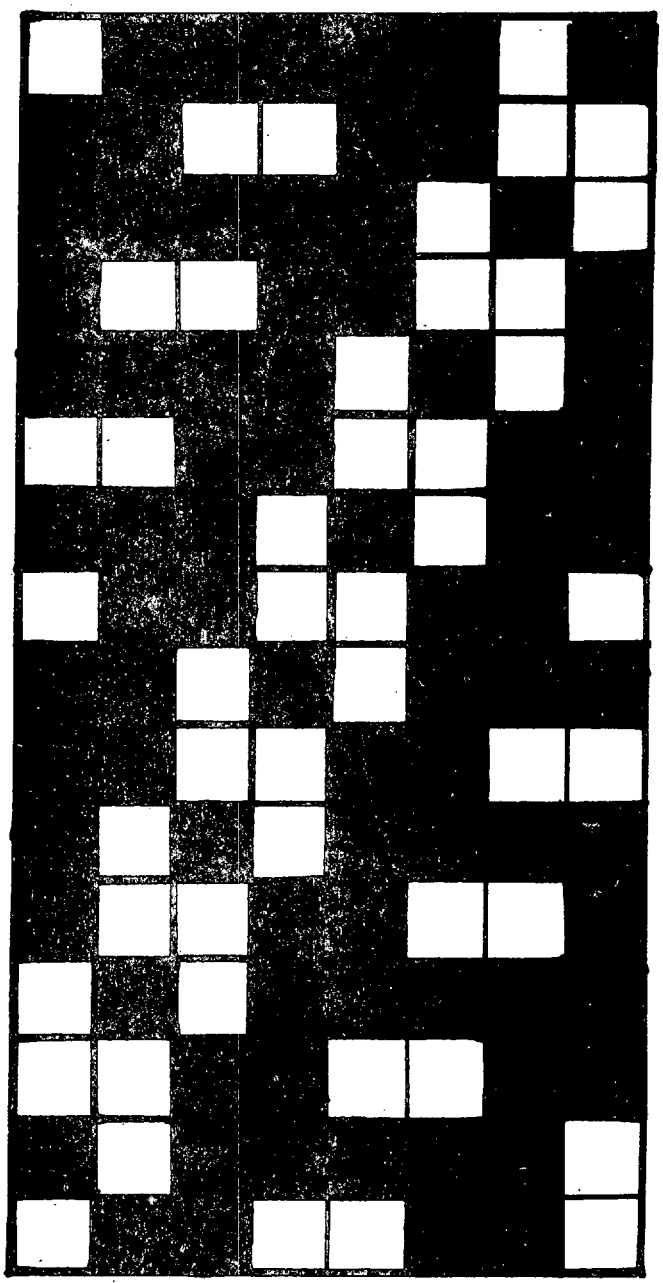


Figure 1 - WEAVE PATTERN

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-24903A(CG)	CLOTH, ELASTIQUE, POLYESTER/WOOL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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